

Curriculum Vitae

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Education

- 2002.9–2006.7 B.S. in Mathematics and Applied Mathematics, Zhengzhou University, Henan Province, P. R. China
- 2006.9–2009.7 M.S. in Applied Mathematics, Graduate School of China Academy of Engineering Physics, Beijing, P. R. China (Advisor: Prof. Changxing Miao)
- 2009.9–2012.7 Ph.D. in Applied Mathematics, Graduate School of China Academy of Engineering Physics, Beijing, P. R. China (Advisor: Prof. Changxing Miao)

Academic Positions

- 2012.9–2013.9 Post-doctor, Laboratoire LAMA, Université Paris-Est, Paris, France (Advisor: Prof. Régis Monneau and Prof. Marco Cannone)
- 2013.10–2016.9 Assistant Professor, Beijing Normal University, Beijing, P. R. China.
- 2016.9–2022.07 Associate Professor, Beijing Normal University, Beijing, P. R. China.
- 2022.8–present Professor, Beijing Normal University, Beijing, P. R. China.
- 2017.2–2017.5 Visiting Researcher, University of California - Irvine, USA.

Research Interests

Nonlinear PDEs, Mathematical Fluid Mechanics.

Research Grants

- National Natural Science Foundation of China, grant No. 11401027, 2015.1–2017.12, Principle Investigator.
- National Natural Science Foundation of China, grant No. 11771043, 2018.1–2021.12, Principle Investigator.
- National Natural Science Foundation of China, grant No. 12271045, 2023.1–2026.12, Principle Investigator.

Publications and Preprints

1. Qianyun Miao, Changhui Tan, Liutang Xue, and Zhilong Xue, Local regularity and finite-time singularity for a class of generalized SQG patches on the half-plane. *ArXiv:2410.19273 [math.AP]*.
2. Omar Lazar, Liutang Xue, and Jiakun Yang, Global regularity and infinite Prandtl number limit of temperature patches for the 2D Boussinesq system. *ArXiv:2405.02137 [math.AP]*.
3. Taoufik Hmidi, Liutang Xue, and Zhilong Xue, Unified theory on V-states structures for active scalar equations. *ArXiv:2312.02874 [math.AP]*.
4. Yatao Li, Qianyun Miao, Changhui Tan, and Liutang Xue, Global well-posedness and refined regularity criterion for the uni-directional Euler-alignment system. To appear in *International Mathematics Research Notices* (2025). <https://doi.org/10.1093/imrn/rnae246>
5. Xiang Bai, Changhui Tan, and Liutang Xue, Global well-posedness and asymptotic behavior for the Euler-alignment system with pressure. *J. Differential Equations*, **407** (2024), 269–310.
6. Xiang Bai, Qianyun Miao, Changhui Tan, and Liutang Xue, Global well-posedness and asymptotic behavior in critical spaces for the compressible Euler system with velocity alignment. *Nonlinearity*, **37** (2024), no. 2, paper no. 025007.
7. Omar Lazar, Yatao Li, and Liutang Xue, On the regularity of temperature fronts for the 3D viscous Boussinesq system. *Indiana Math. Univ. J.*, **73** (2024), no. 3, 1005–1060.
8. Taoufik Hmidi, Liutang Xue, and Zhilong Xue, Emergence of time periodic solutions for the generalized surface quasi-geostrophic equation in the disc. *Journal of Functional Analysis*, **285** (2023), no. 10, Paper No. 110142.
9. Yatao Li, Qianyun Miao, and Liutang Xue, Global well-posedness for 2D fractional inhomogeneous Navier-Stokes equations with rough density. *Nonlinearity*, **36** (2023), 3866–3908.
10. Dongho Chae, Qianyun Miao, and Liutang Xue, Global regularity of non-diffusive temperature fronts for the 2D viscous Boussinesq system. *SIAM J. Math. Anal.*, **54** (2022), no. 4, 4043–4103.
11. Piotr B. Mucha, and Liutang Xue, Boussinesq system with measure forcing. *Mathematische Annalen*, **382** (2022), 1901–1934.
12. Qianyun Miao, Changhui Tan, and Liutang Xue, Global regularity for a 1D Euler-alignment system with misalignment. *Math. Mod. Meth. Appl. Sci.*, **31** (2021), no. 3, 473–524.
13. Qianyun Miao, and Liutang Xue, Regularity and singularity results for the dissipative Whitham equation and related surface wave equations. *Commun. Math. Soc.*, **17** (2019), no. 8, 2141–2190.
14. Changxing Miao, and Liutang Xue, On the regularity issues of a class of drift-diffusion equations with nonlocal diffusion. *SIAM J. Math. Anal.*, **51** (2019), no. 4, 2927–2970.
15. Omar Lazar, and Liutang Xue, Regularity results for a class of generalized surface quasi-geostrophic equations. *Journal de Mathématiques Pures et Appliquées*, **130** (2019), 200–250.
16. Piotr B. Mucha, Liutang Xue, and Xiaoxin Zheng, Between homogeneous and inhomogeneous Navier-Stokes systems: the issue of stability. *J. Differential Equations*, **267** (2019), Issue 1, 307–363.
17. Qianyun Miao, and Liutang Xue, On the Hölder regularity of the weak solution to a drift-diffusion equation with pressure. *Calc. Var. Partial Differential Equations*, **57** (2018), no. 6, Art. 153, 33 pp.
18. Liutang Xue, and Zhuan Ye, On the differentiability issue of the drift-diffusion equation with nonlocal Lévy-type diffusion. *Pacific J. Mathematics*, **293** (2018), no. 2, 471–510.

19. Zhuan Ye, Xiaojing Xu, and Liutang Xue, On the global regularity of the 2D Boussinesq equations with fractional dissipation. *Math. Nachr.*, **290** (2017), no. 8-9, 1420–1439.
20. Jiahong Wu, Xiaojing Xu, Liutang Xue, and Zhuan Ye, Regularity results for the 2D Boussinesq equations with critical or supercritical dissipation. *Commun. Math. Sci.*, **7** (2016), 1963–1997.
21. Liutang Xue, On the locally self-similar solution of the surface quasi-geostrophic equation with decaying or non-decaying profiles. *J. Differential Equations*, **261** (2016), no. 10, 5590–5608.
22. Liutang Xue, On the locally self-similar singular solutions for the incompressible Euler equations. *Dyn. Partial Differ. Equ.*, **12** (2015), no. 4, 321–342.
23. Liutang Xue, On the discretely self-similar solutions for the 3D Navier-Stokes equations. *Nonlinearity*, **28** (2015), no. 10, 3695–3708.
24. Marco Cannone, and Liutang Xue, Remarks on self-similar solutions for the surface quasi-geostrophic equation and its generalization. *Proc. Amer. Math. Soc.*, **143** (2015), no. 6, 2613–2622.
25. Dong Li, Changxing Miao, and Liutang Xue, On the well-posedness of a 2D nonlinear and nonlocal system arising from the dislocation dynamics. *Commun. Contemp. Math.*, **16** (2014), no. 2, 1350021, 36 pp.
26. Xiaojing Xu, and Liutang Xue, Yudovich type solution for the 2D inviscid Boussinesq system with critical and supercritical dissipation. *J. Differential Equations*, **256** (2014), no. 9, 3179–3207.
27. Marco Cannone, Changxing Miao, and Liutang Xue, Global regularity for the supercritical dissipative quasi-geostrophic equation with large dispersive forcing. *Proc. Lond. Math. Soc.*, **106** (2013), no. 3, 650–674.
28. Liutang Xue, and Xiaoxin Zheng, Note on the well-posedness of a slightly supercritical surface quasi-geostrophic equation. *J. Differential Equations*, **253** (2012), 795–813.
29. Gang Wu, and Liutang Xue, Global well-posedness for the 2D inviscid Bénard system with fractional diffusivity and Yudovich's type data. *J. Differential Equations*, **253** (2012), 100–125.
30. Changxing Miao, and Liutang Xue, Global well-posedness for a modified critical dissipative quasi-geostrophic equation. *J. Differential Equations*, **252** (2012), 792–818.
31. Changxing Miao, and Liutang Xue, On the regularity of a class of generalized quasi-geostrophic equations. *J. Differential Equations*, **251** (2011), 2789–2821.
32. Changxing Miao, and Liutang Xue, On the global well-posedness of a class of Boussinesq-Navier-Stokes systems. *Nonlinear Differential Equations and Applications*, **18** (2011), 707–735.
33. Liutang Xue, Wellposedness and zero microrotation viscosity limit of the 2D micropolar fluid equations. *Mathematical Methods in the Applied Sciences*, **34** (2011), 1760–1777.
34. Liutang Xue, On the well-posedness of incompressible flow in porous media with supercritical diffusion. *Applicable Analysis*, **88** (2009), no. 4, 547–561.

Invited Talks (Partially)

- Beijing conference on harmonic analysis and partial difference equations. IAPCM, Beijing, 17 May–23 May 2010.

- Dynamics of nonlinear dispersive and fluid mechanics equations, BICMR, Beijing, 25 June–13 July 2012.
- PDE Seminar, Rennes 1 University, France, 10 January 2013.
- The Fifth BNU-PDE Workshop, BNU, Beijing, 24 January–25 January 2015.
- PDE Seminar, Chung-Ang University, South Korea, 10 May 2016.
- PDE Seminar, University of California - Irvine, USA, 11 April 2017.
- Workshop Mathflows 2018, Porquerolles, France, 3 September 2018.
- Workshop on Mathematical Theory of Fluid Mechanics, BNU, Beijing, 6 July 2019.
- Workshop Mathflows 2020, Bedlewo, Poland, 8-13 March 2020. (cancelled)
- The 7th PDEs Youth Scholars Forum, South China Normal University, Guangzhou, China, 28-30 November 2020.
- Recent Developments in the Nonlinear PDEs in Fluid Mechanics and Related Fields, Zhejiang Normal University, Jinhua, China, 16-17 October 2021.
- International Workshop on Multiphase Flows: Analysis, Modelling and Numerics, Waseda University, Japan, 30 Nov. - 3 Dec. 2021. (online)
- Fluid Conversation Seminar, University of Seville, Spain, 3 March 2022. (online)
- Analysis & PDE Seminar, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, China 30 September 2022.
- Partial Differential Equations Theory and Applications, Central China Normal University, China, 10-11 Dec. 2022. (online)
- Workshop on the PDEs in Fluid Mechanics, Zhengzhou University of Light Industry, China, 15-16 July 2023.
- Mathematical Theory of Fluid Equations and Related Models, Beihang University, China, 30-31 Dec. 2023.
- KinMAT Workshop, Warsaw, Poland, 3-7 June 2024.
- Alhambra PDE days (Satellite Conference of the 9th European Congress of Mathematics), University of Granada, Spain, 8-12 July 2024.
- The 14th AIMS Conference, NYU Abu Dhabi, Abu Dhabi, UAE, 16-20 Dec. 2024.